

澳門科技大學
MACAU UNIVERSITY OF SCIENCE AND TECHNOLOGY

Text-driven image style transfer: Achieving precise visual manipulation through stable diffusion, PTP, and NTI

Zhang Xiting, Guo Jiawei,
Wang Xiaoye, Lu Siyuan

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01 Background & Significance

02 Dataset

03 Methods

04 Model Evaluation

05 Task Division



Input



Others' Output



Our Goal

In recent years, the advancements in generative models like Stable Diffusion have revolutionized how we create images based on textual descriptions. However, the challenge remains in editing existing images or real photographs with the same level of precision and control that these models offer for image generation.

Current Limitations:
Traditional methods such as GANs often fail to maintain the original structure and details when making edits (e.g., changing a red apple to a green one might alter its shape or position).

Our Goal:
To develop a pipeline that allows for precise modifications of both generated and real-world images using text prompts, ensuring high fidelity in content preservation while enabling flexible style adjustments.

Significance:
This research contributes significantly by:
Offering a novel solution for controlled image editing, which can be applied in various fields including digital art, advertising, and personalized content creation.
Providing a robust framework for future innovations in text-driven image editing.

COCO val2017 Dataset + real photos

1. Scenery
2. Food
3. Street View
4. Human Faces
5. Animals
6. Indoor scenes







Methods

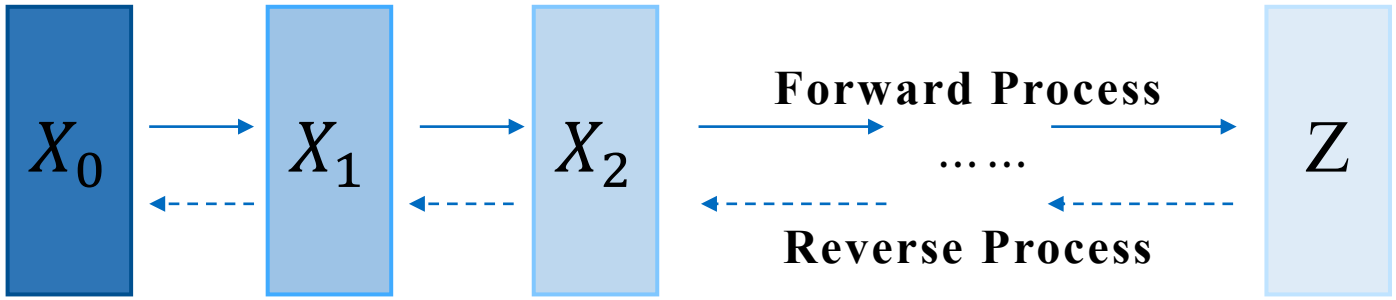
01 Diffusion Model

02 Prompt-to-Prompt

03 Null Text Inversion

➢ What is Diffusion Model – It's core principles

Diffusion Model: Gradually add Gaussian noise and then reverse



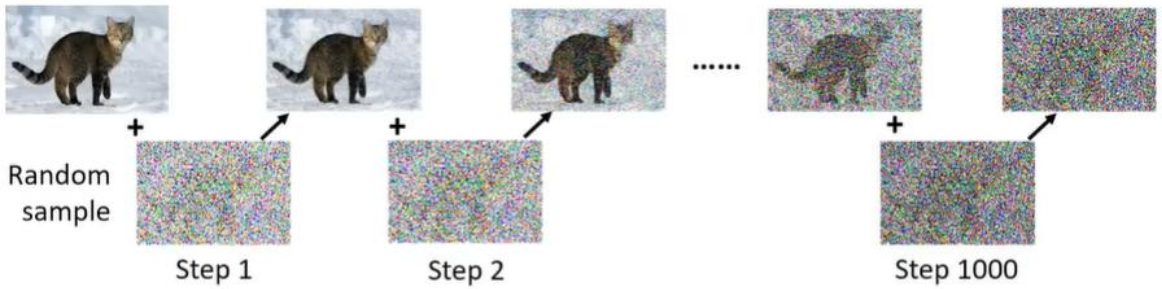
Forward Process

Starting from the real image, gradually add noise until the image completely becomes random noise.

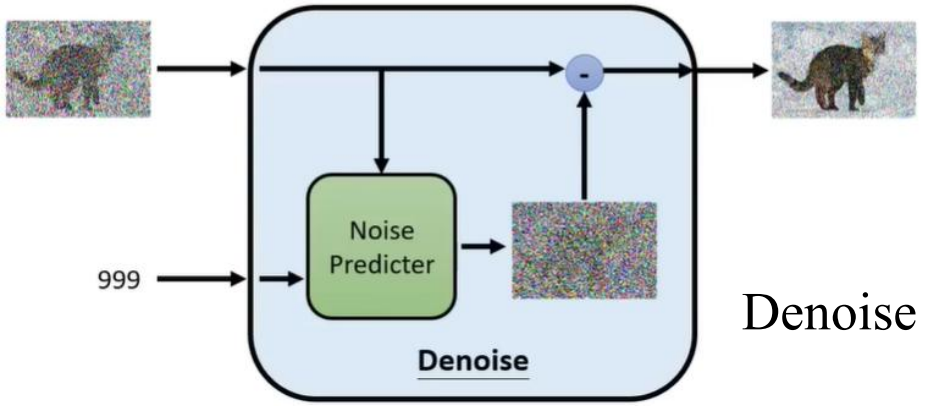
Reverse Process

Start from noisy images and gradually restore clear images.

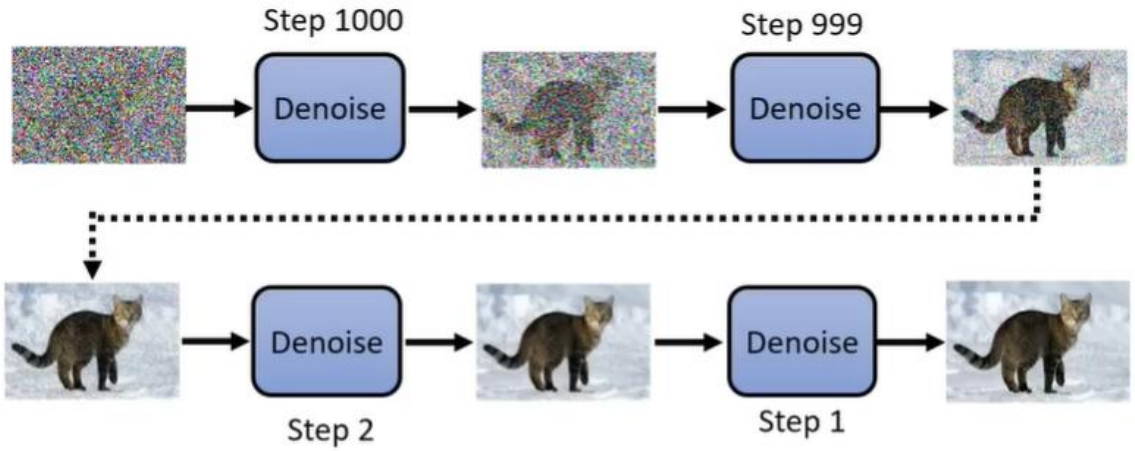
➢ What is Diffusion Model – How to Denoise



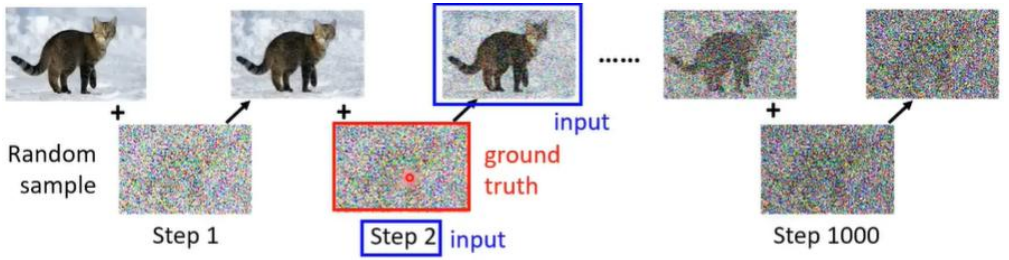
Forward Process

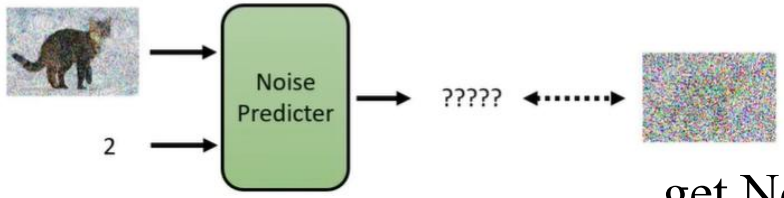


Denoise Module do



Reverse Process

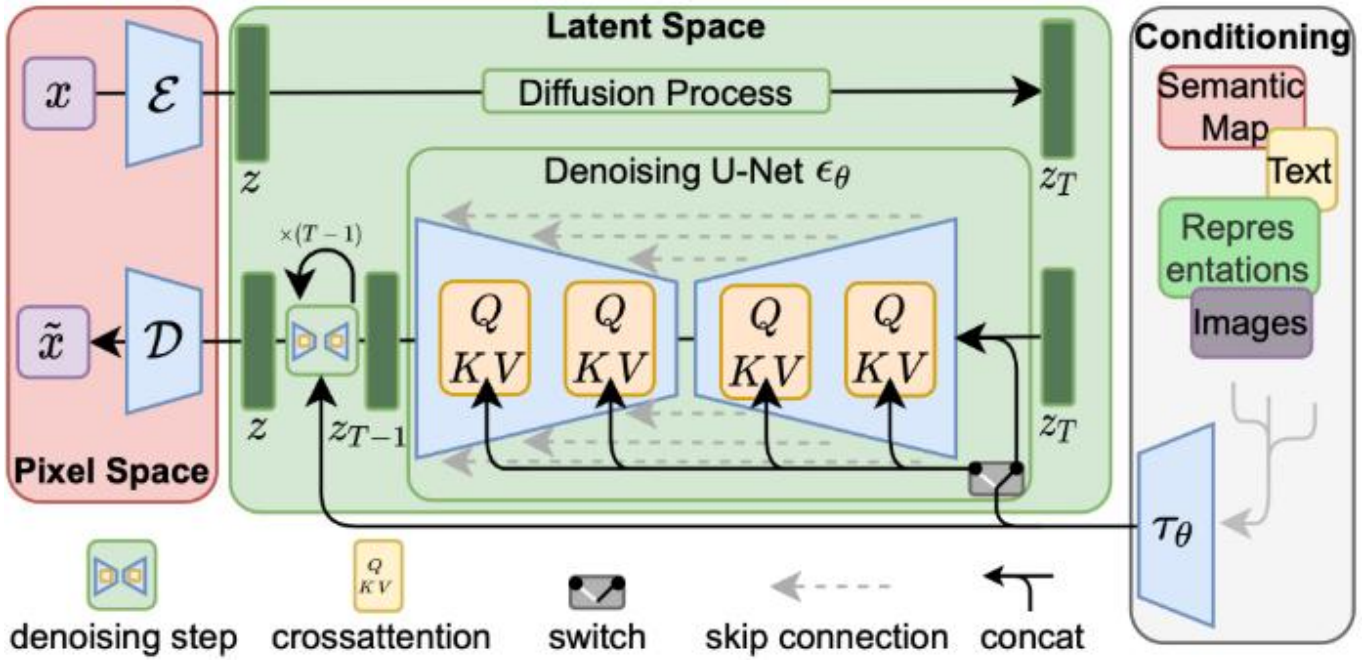




get Noise Predicter

扩散模型 - Diffusion Model【李宏毅2023】_哔哩哔哩_bilibili

➢ Stable Diffusion – Framework and Three Core Modules



Latent Space

The diffusion process actually occurs.

The core component is Denoising U-Net, which gradually restores the noise z_T to a clear z through multiple denoising steps.

The **Q, K, V** represent the Cross-Attention mechanism. Query comes from the image features, Key and Value come from the conditional we input.

Pixel Space

Encoder ε and Decoder D.

The input image x is here compressed by the encoder to Latent representation z, or restored by the decoder to image $\tilde{x}$

Conditional control

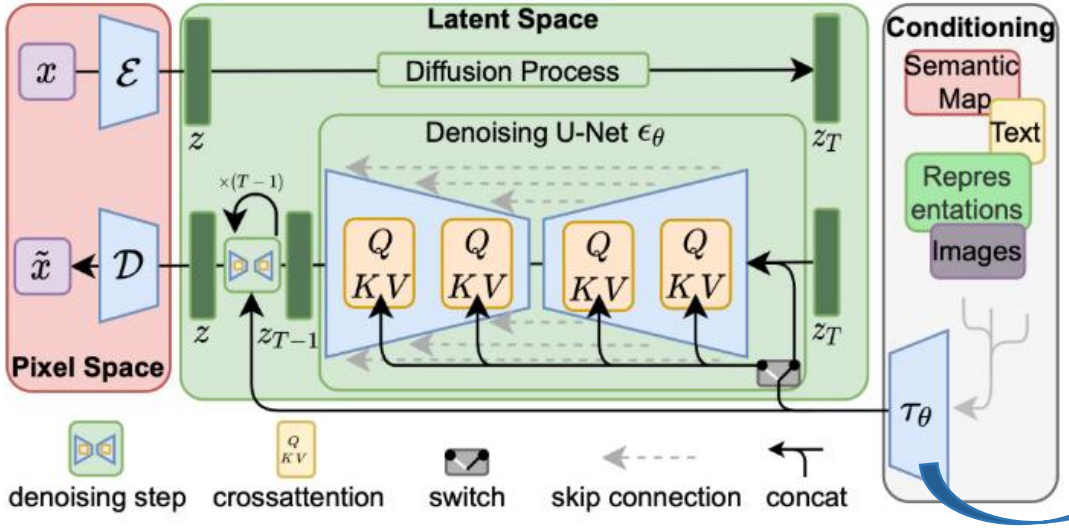
supports multiple conditions: text (encoded by CLIP), semantic segmentation images…

These conditions are transformed into feature representations through the encoder τ_θ and injected into the Cross-Attention layer of U-Net.

[2112.10752] High-Resolution Image Synthesis with Latent Diffusion Models

➢ How to achieve style transfer

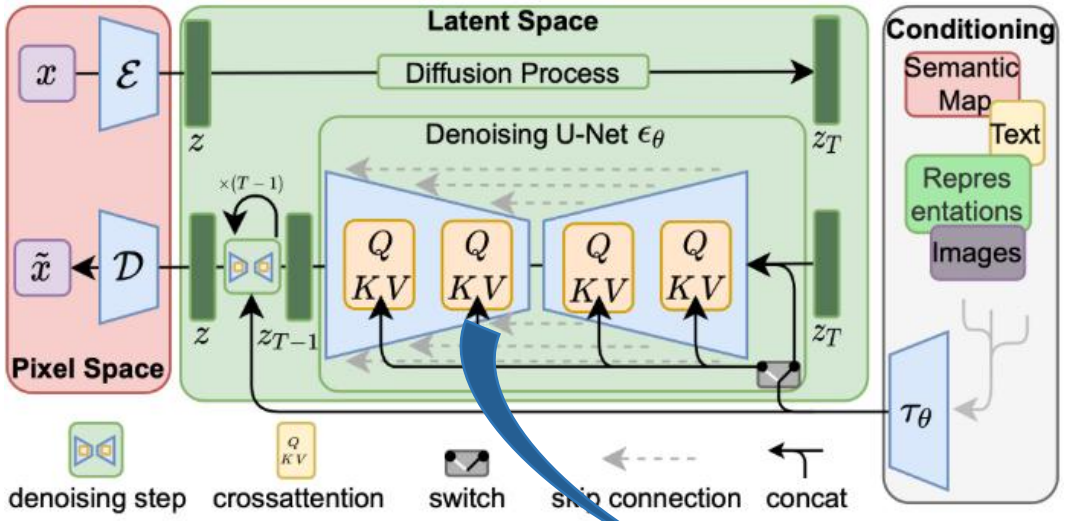
The core idea of style transfer based on the diffusion model is the inversion - redrawing strategy: First, the content image is inverted into noise through some methods, and this noise encodes the structural information of the original image; Then, denoise with style prompt words to enable the model to generate in the new style while maintaining its structure.



Style instruction encoding:
Take the "style description" as the Text input, which is translated into a numerical vector recognizable by the model by the CLIP text encoder $\boldsymbol{\tau_\theta}$, and then enters the Conditioning module as the "style instruction" for the denoising process.

[2112.10752] High-Resolution Image Synthesis with Latent Diffusion Models

➢ How to achieve style transfer



Conditional guided denoising:

While denoising and restoring the content structure, the target style is added to the corresponding position of the image.

Cross-Attention (Q/K/V) fusion style condition:

U-Net extracts image features from the latent variable z_T as Query (***Q***);

Style text representation is input as Key (***K***) and Value (***V***) for Cross-Attention;

Cross-Attention calculates the attention weights of ***Q*** and ***K***, injects style information (***V***) into the image features, and enables the model to know which style to apply in which areas.

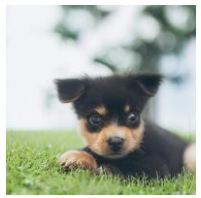
[2112.10752] High-Resolution Image Synthesis with Latent Diffusion Models

➢ Limitations and What we expect

01 Image to Image

When noise is added to the image for redrawing, the content and style are mixed together and cannot be precisely separated, resulting in the loss of some information








02 DDIM inversion

Reversible diffusion is achieved by using deterministic sampling, but the prediction error of the model will accumulate, resulting in imperfect reconstruction.

Optimize inversion to achieve precise reconstruction

Null-text Inversion

Operate the attention map to achieve precise style control

Prompt to prompt

Pixel-level content retention and semantic-level style control

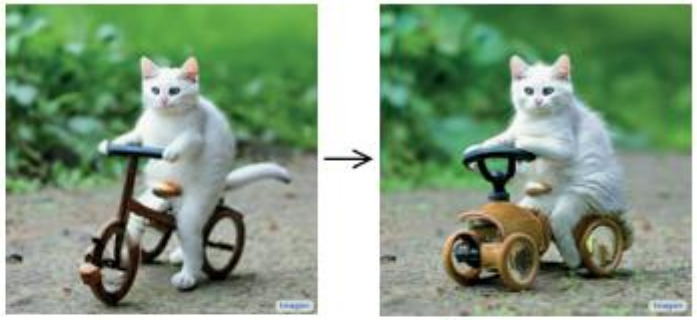
DDIM 反演 - Hugging Face 扩散模型课程 - Hugging Face 文档

➢ Prompt-to-Prompt

An intuitive **image editing** technique, of manipulating local or global details by **modifying only the text prompt** when using text-guided diffusion models. By **injecting internal cross-attention maps**, they **preserve the spatial layout and geometry** which enable the regeneration of an image while modifying it through prompt editing.



"The boulevards are crowded today."



"Photo of a cat riding on a bicycle."



"Children drawing of a castle next to a river."



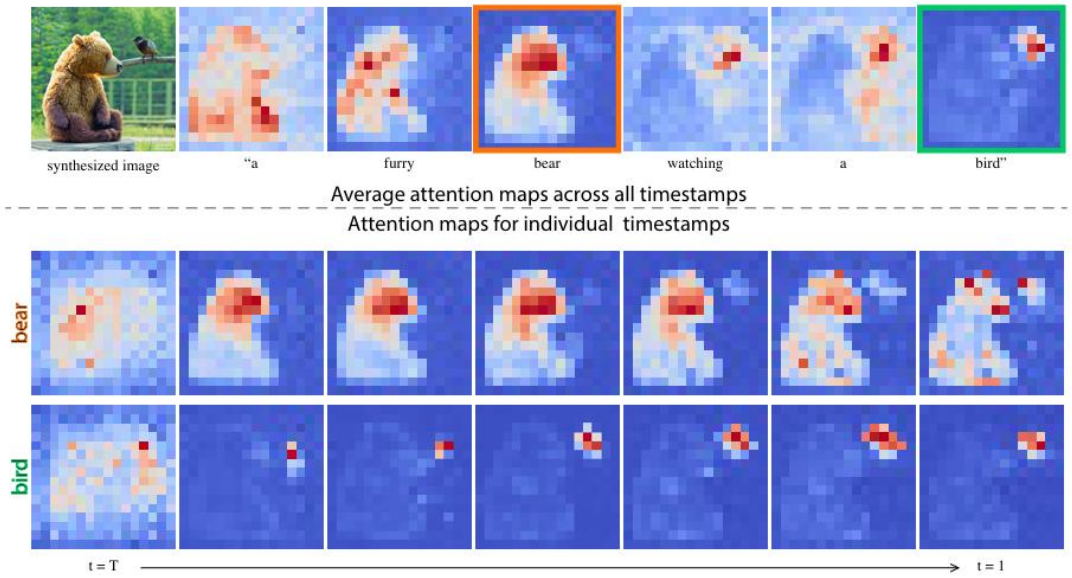
"a cake with decorations."

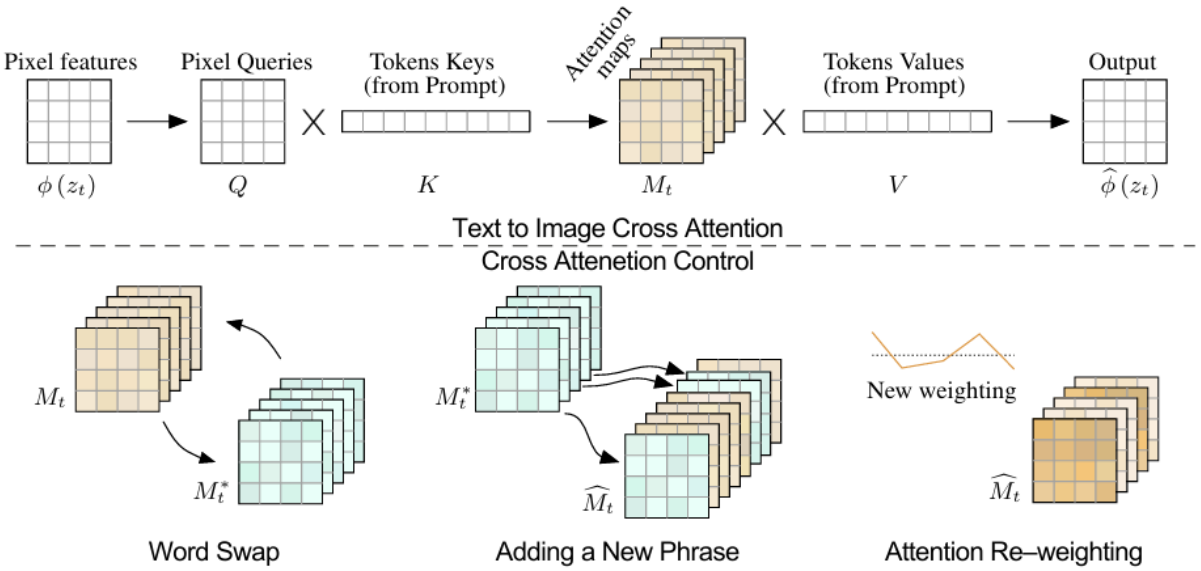
Prompt-to-Prompt primarily offers the following **three editing operations:**

- **Word Swap:** The user swaps tokens of the original prompt with others, e.g., "*a big red bicycle*" → "*a big red car*".
- **Adding a New Phrase:** The user adds new tokens to the prompt, e.g., "*a castle next to a river*" → "*children drawing of a castle next to a river*".
- **Attention Re–weighting:** The user may wish to strengthen or weakens the extent to which each token is affecting the resulting image. e.g., the prompt "*a fluffy red ball*" → we want to make the ball *more or less* fluffy.

➢ *Prompt-to-Prompt:* Controlling the Cross-attention

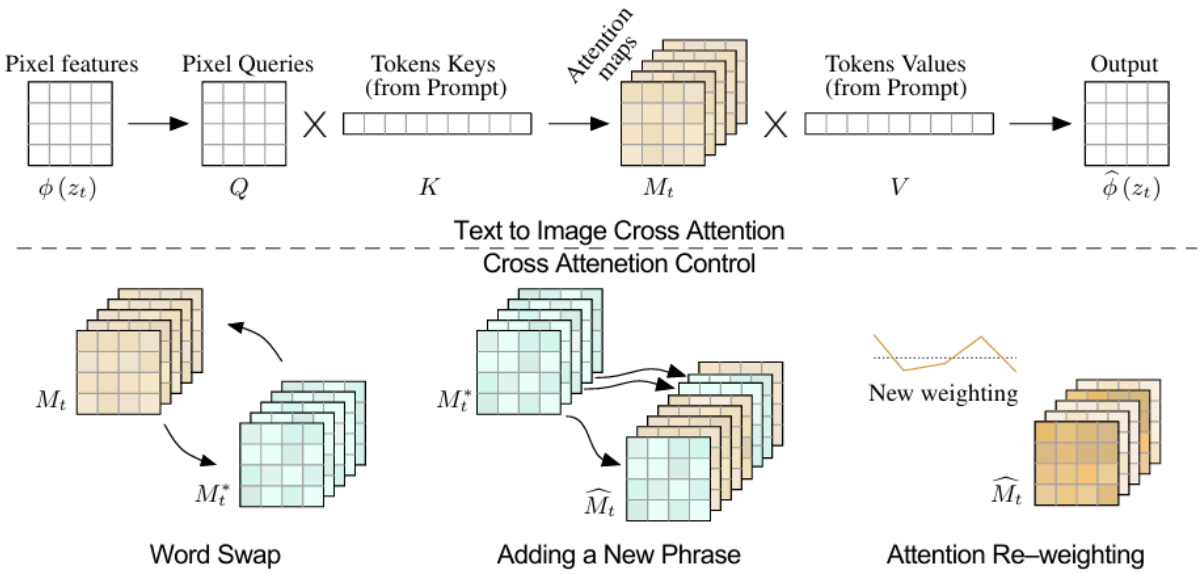
The ***cross-attention layers*** are the key to **controlling the relation between the spatial layout of the image to each word in the prompt**. And ***cross-attention maps***, which are high-dimensional tensors, can **bind pixels and tokens extracted from the prompt text.** Therefore, we can edit images by injecting the cross-attention maps during the diffusion process, controlling which pixels attend to which tokens of the prompt text during which diffusion steps. We control the spatial layout and geometry of the generated image using the attention maps of a source image.



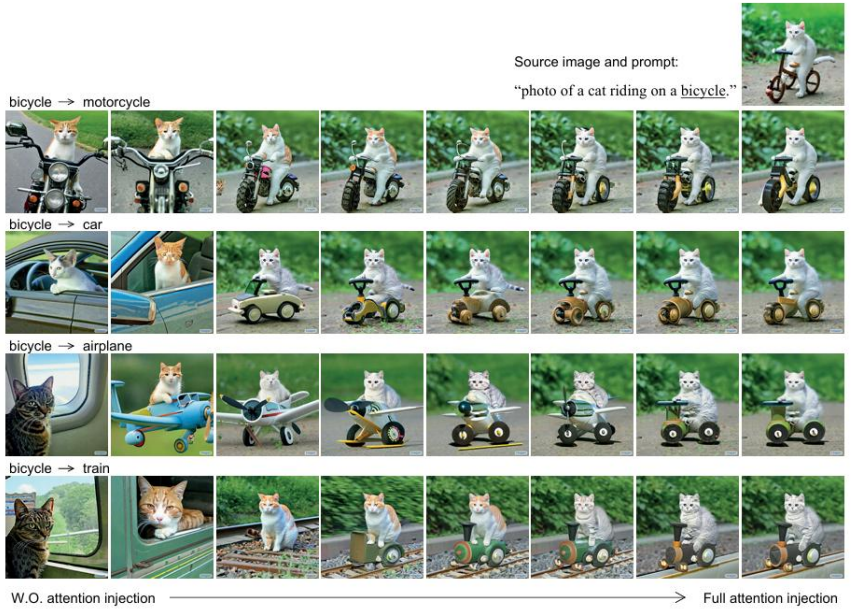


➢ *Prompt-to-Prompt:* Controlling the Cross-attention

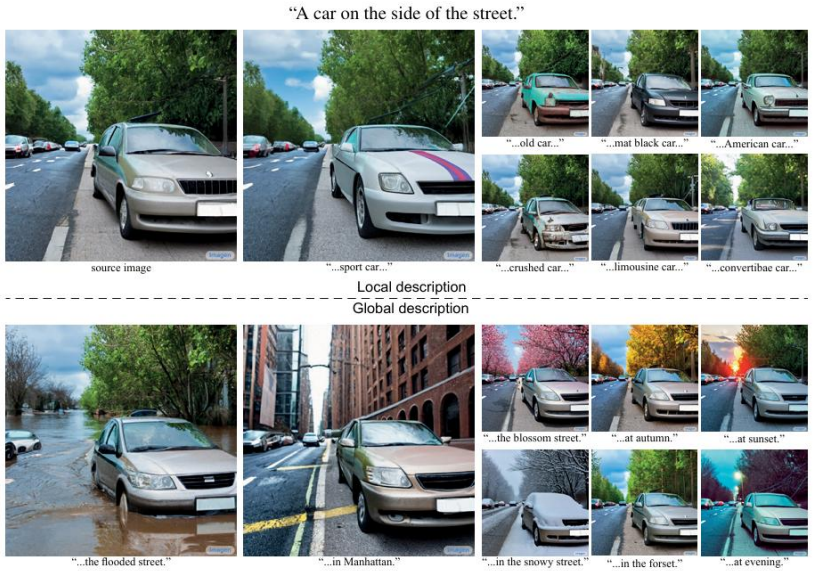
- **Word Swap:** When swapping a word in the prompt, we inject the source image maps M_t, overriding the target image maps M_t^*, to preserve the spatial layout.
- **Adding a New Phrase:** Inject only the maps that correspond to the unchanged part of the prompt, which enables editing of image details or styles while preserving the original image's spatial layout.
- **Attention Re–weighting:** To amplify or attenuate the semantic effect of a word, simply re-weighting the corresponding attention map.



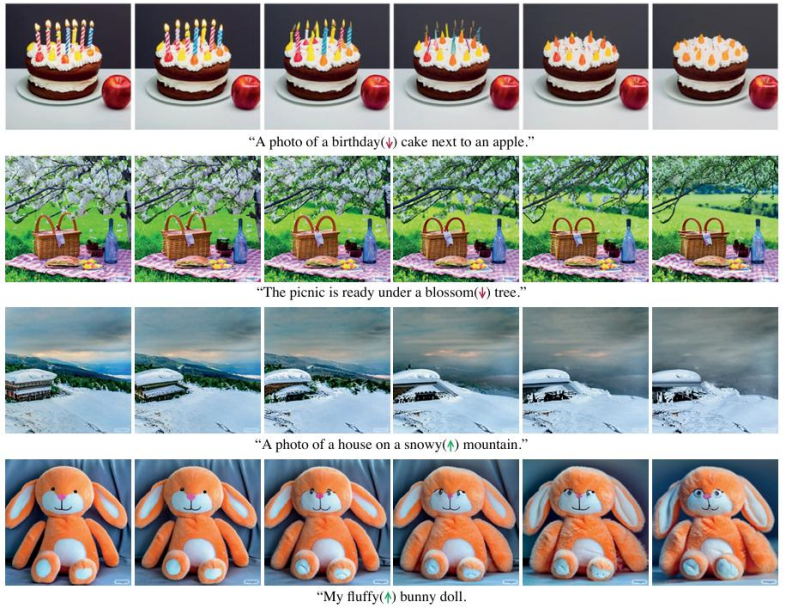
Controlling the Cross-attention



Word Swap



Add a New Phrase



Attention Re–weighting



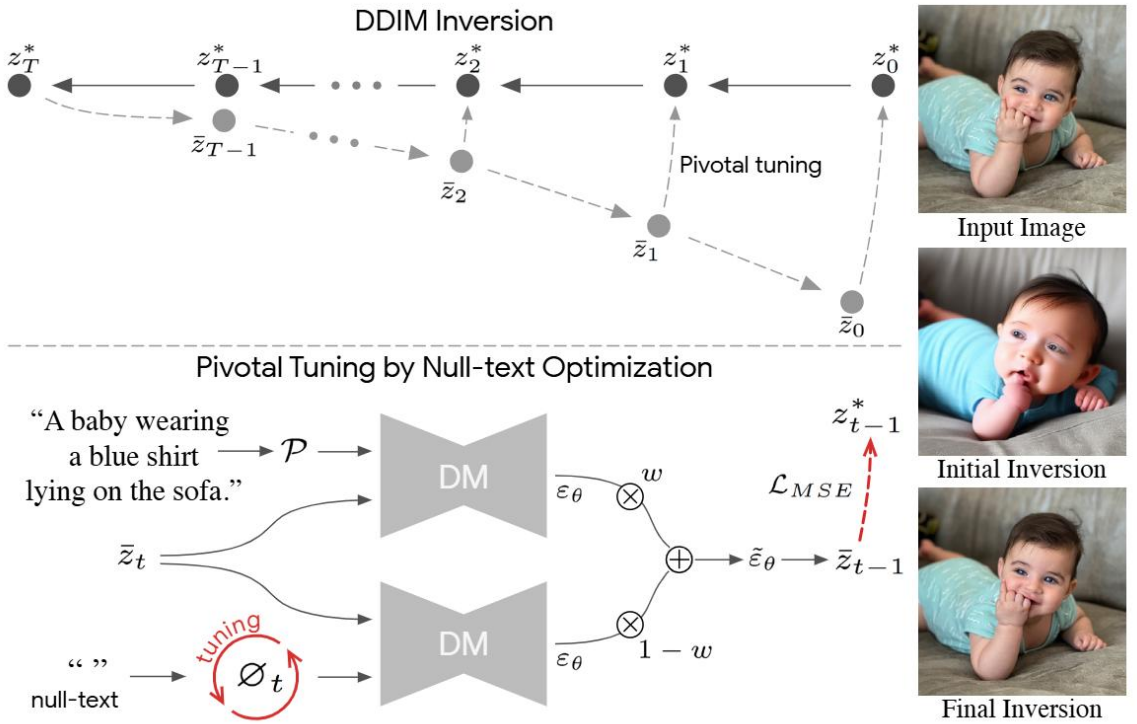
Style Transfer

Prompt-to-Prompt does not require model training, fine-tuning, extra data, or optimization and can be deployed immediately for editing tasks.

However, without an image inversion technique, their approach is limited to synthesized images.

➢ Null-text Inversion

Null-text Inversion is an image reconstruction and text-driven editing method. It embeds real images into the latent space of diffusion models through two key steps: pivotal inversion and null-text optimization, enabling high-fidelity reconstruction and text-driven editing.



Two Core of Null-text Inversion——*Pivotal Inversion* and *Null-text Optimization*

Pivotal Inversion: The initial trajectory obtained by DDIM inversion is used as the "hub", and the optimization around it is carried out to gradually approach the original image coding, avoiding random noise initialization, improving efficiency and editability.

Null-text Optimization: Only the null-text embedding is optimized, while the model weights and conditional embedding remain unchanged, thereby preserving the model's editing capabilities.

➢ Core Capabilities of NTI

The core contribution of NTI lies in being the first method that enables the Prompt-to-Prompt technique to be applied to real image editing. Through pivotal inversion and null-text optimization, NTI achieves intuitive text-driven modifications while maintaining high fidelity. Its specific capabilities include:

- **High-fidelity Reconstruction**

By optimizing the null-text embedding, NTI can almost perfectly reconstruct the input real image, avoiding the distortions of traditional DDIM inversion under high guidance scales.

- **Text-guided Editing**

After completing the inversion, the user only needs to modify the prompt to perform editing. For example,:

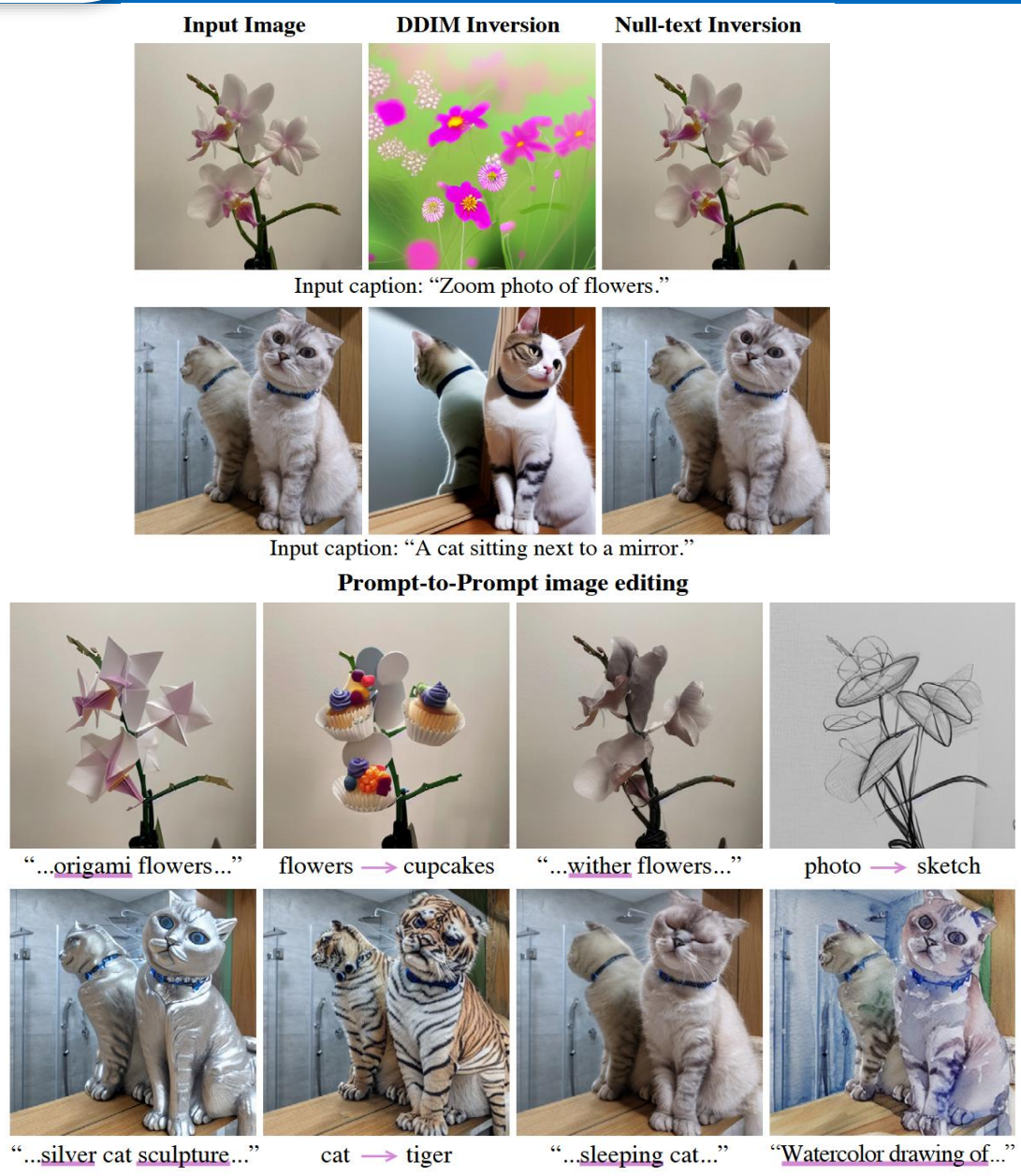
“cat → tiger” （Object Swap）

“photo → sketch” （Style Change）

“flowers → cupcakes” （Semantic Replacement）

- **Multiple Edits**

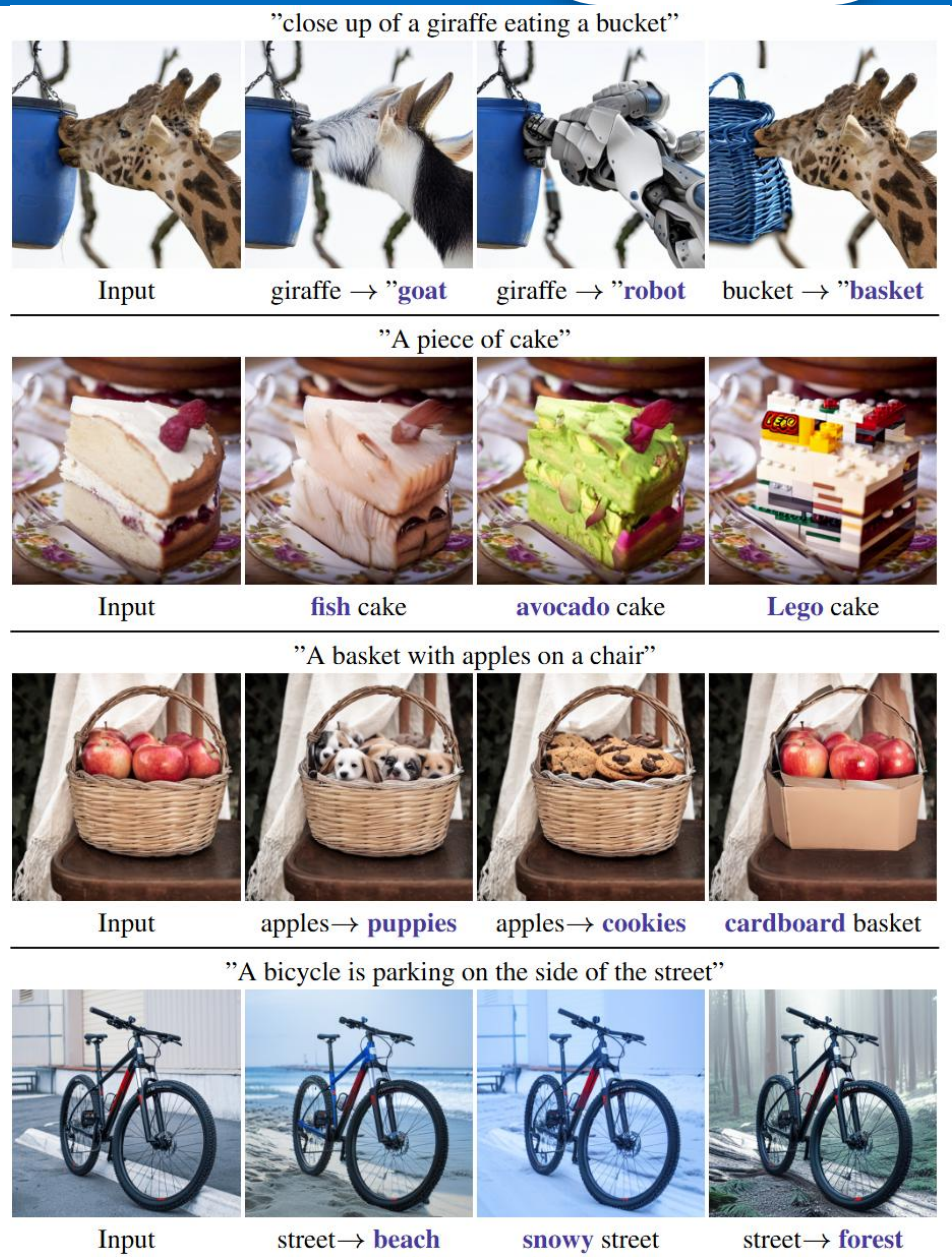
The optimized embedding obtained after a single inversion can be reused, supporting multiple different Prompt-to-Prompt edits on the same image without the need for retraining or fine-tuning the model.



Editing Examples



NTI retains the strengths of Prompt-to-Prompt while extending its applicability from synthesized images to real image editing.



DINO

- higher = better
- Determine whether the position, posture and layout of the object after editing are consistent with the original image

LPIPS

- lower = better
- Judge whether the details are natural and whether there is any blurring or artifacts

CLIP

- higher = better
- Determine whether the generated image conforms to the semantics of the new prompt words (such as "ink wash painting", "Van Gogh style")

Member A

- Data Collection and Cleaning
- Implement quantitative and qualitative assessment plans

Member B

- Focus on diffusion model implementation and optimization.

Member C

- Lead the development and testing of PTP functionalities.

Member D

- Specialize in NTI technique application and improvement.

Work Plan

Weeks 1–2

01

Data Collection and Cleaning, and setup of the Stable Diffusion baseline pipeline.

Weeks 3–4

02

Implement Prompt-to-Prompt (PtP) and validate its ability to preserve structural consistency during editing.

Weeks 5–6

03

Implement Null-Text Inversion (NTI) and evaluate reconstruction fidelity on real images.

Weeks 7

04

Integrate NTI + PtP and conduct preliminary style transfer experiments. Perform ablation studies, baseline comparisons, and user study (optional).

Weeks 8

05

Finalize results, write project report, and prepare presentation slides.

Reference

[1] *Denoising Diffusion Probabilistic Models*, J. Ho et al., **NeurIPS 2020**
[2] *High-Resolution Image Synthesis with Latent Diffusion Models*, R. Rombach et al., **CVPR 2022**
[3] *Learning Transferable Visual Models From Natural Language Supervision*, A. Radford et al., **ICML 2021**
[4] *Denoising Diffusion Implicit Models*, J. Song et al., **ICLR 2021**
[5] *Prompt-to-Prompt Image Editing with Cross-Attention Control*, A. Hertz et al., **ICLR 2023**
[6] *Null-text Inversion for Editing Real Images using Guided Diffusion Models*, R. Mokady et al., **CVPR 2023**
[7] *Guided Image Synthesis and Editing with Stochastic Differential Equations*, C. Meng et al., **ICLR 2022**
[8] *An Edit Friendly DDIM Inversion for Generating High-Resolution Images*, M. Panashchuk et al., **arXiv 2023**



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THANK YOU

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